Tackling Noise with Error Correction

Quantum AI (417523A) - SPPU BE AI&DS

CODE

```
import warnings

from qiskit import QuantumCircuit, QuantumRegister, ClassicalRegister, transpile
from qiskit_aer import AerSimulator

warnings.filterwarnings("ignore", category=DeprecationWarning)

SHOTS = 2048


def repetition_code(correct):
    data = QuantumRegister(3, "data")
    anc = QuantumRegister(2, "anc")
    syn = ClassicalRegister(2, "syn")
    out = ClassicalRegister(3, "out")
    qc = QuantumCircuit(data, anc, syn, out)

    qc.x(data[0])
    qc.cx(data[0], data[1])
    qc.cx(data[0], data[2])

    qc.x(data[1])

    if correct:
        qc.cx(data[0], anc[0])
        qc.cx(data[1], anc[0])
        qc.cx(data[0], anc[1])
        qc.cx(data[2], anc[1])
        qc.measure(anc[0], syn[0])
        qc.measure(anc[1], syn[1])
        with qc.if_test((syn, 1)):
            qc.x(data[1])
        with qc.if_test((syn, 2)):
            qc.x(data[2])
        with qc.if_test((syn, 3)):
            qc.x(data[0])

    qc.measure(data, out)
    return qc


print("=" * 66)
print("  QUANTUM AI (417523A)  -  PRACTICAL 4")
print("  TACKLING NOISE WITH ERROR CORRECTION  (3-qubit bit-flip code)")
print("=" * 66)

print("\nLogical |1> is encoded as |111> across three physical qubits.")
print("A bit-flip error is then forced on the middle qubit, so the")
print("register holds |101> while the syndrome detects the damage.")

sim = AerSimulator()
# from qiskit_aer.noise import NoiseModel, depolarizing_error
# sim = AerSimulator(noise_model=NoiseModel().add_all_qubit_quantum_error(depolarizing_error(0.05, 1),
["x"]))

print("\n[Experiment A]  Error introduced, error correction DISABLED")
raw = repetition_code(correct=False)
raw_counts = sim.run(transpile(raw, sim), shots=SHOTS).result().get_counts()
print("   codeword (data2 data1 data0) counts over %d shots:" % SHOTS)
for key, count in raw_counts.items():
    print(f"      codeword = {key.split()[0]} : {count:5d}  ({100 * count / SHOTS:5.2f} %)")
print("   The codeword |111> has become |101>: the protected register")
print("   is damaged and no longer carries a valid copy of the qubit.")
```

```
print("\n[Experiment B]  Error introduced, error correction ENABLED")
fixed = repetition_code(correct=True)
fixed_counts = sim.run(transpile(fixed, sim), shots=SHOTS).result().get_counts()

print("   syndrome value 1  -> flip data[1]   (parity of data0,data1 broken)")
print("   syndrome value 2  -> flip data[2]   (parity of data0,data2 broken)")
print("   syndrome value 3  -> flip data[0]   (both parities broken)")
print("\n   (codeword, syndrome) counts over %d shots:" % SHOTS)
for key, count in sorted(fixed_counts.items(), key=lambda kv: -kv[1]):
    print(f"      codeword={key.split()[0]}  syndrome={key.split()[1]} : {count:5d}  ({100 * count /
SHOTS:5.2f} %)")

print("\n   The syndrome is always 01, so the correction circuit knows the")
print("   flip happened on data[1] and applies X through the conditional.")
print("   The codeword is restored to |111> and the logical qubit read")
print("   out is 1 in 100 % of the shots. Error correction succeeded.")
print("=" * 66)
```

OUTPUT

```
==================================================================
  QUANTUM AI (417523A)  -  PRACTICAL 4
  TACKLING NOISE WITH ERROR CORRECTION  (3-qubit bit-flip code)
==================================================================

Logical |1> is encoded as |111> across three physical qubits.
A bit-flip error is then forced on the middle qubit, so the
register holds |101> while the syndrome detects the damage.

[Experiment A]  Error introduced, error correction DISABLED
   codeword (data2 data1 data0) counts over 2048 shots:
      codeword = 101 :  2048  (100.00 %)
   The codeword |111> has become |101>: the protected register
   is damaged and no longer carries a valid copy of the qubit.

[Experiment B]  Error introduced, error correction ENABLED
   syndrome value 1  -> flip data[1]   (parity of data0,data1 broken)
   syndrome value 2  -> flip data[2]   (parity of data0,data2 broken)
   syndrome value 3  -> flip data[0]   (both parities broken)

   (codeword, syndrome) counts over 2048 shots:
      codeword=111  syndrome=01 :  2048  (100.00 %)

   The syndrome is always 01, so the correction circuit knows the
   flip happened on data[1] and applies X through the conditional.
   The codeword is restored to |111> and the logical qubit read
   out is 1 in 100 % of the shots. Error correction succeeded.
==================================================================
```

FIGURE: Q4_bitflip_error_correction

